

Ryan Zeyuan Chen

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🌐 <https://ryan-zeyuan-chen.github.io/>

☎ 905-617-5497

📍 Toronto, ON, Canada

Education

Toronto, ON, Canada
Sep 2022 – Apr 2025

M.A.Sc. in Mechanical Engineering, University of Toronto
Advisor: Prof. Andrew Goldenberg
Collaborative Specialization in Robotics

Toronto, ON, Canada
Sep 2017 – Apr 2022

B.A.Sc. in Engineering Science, University of Toronto
Advisor: Prof. Jessica Burgner-Kahrs
Robotics Engineering Major, Artificial Intelligence Engineering Minor, Engineering Business Certificate

Research Experience

Toronto, ON, Canada
Sep 2022 – Apr 2025

Robotics Institute, University of Toronto | Graduate Researcher
Supervised by Prof. Andrew Goldenberg - Robotics and Automation Laboratory

- Designed a transfer learning framework to improve data efficiency and generalizability in learning-based kinematics modeling for robots.
- Applied the framework to concentric tube continuum robots (CTCRs) via PyTorch, reducing the amount of real-world training data needed by 90% for forward kinematics and 33% for inverse kinematics.

Boston, MA, USA
May 2023 – Sep 2023

Brigham and Women's Hospital, Harvard Medical School | Robotics Research Intern
Supervised by Prof. Jayender Jagadeesan - Surgical Imaging and Mechatronics Laboratory

- Developed a novel dynamics model for tendon-driven continuum robots (TDCRs) by incorporating static and sliding tendon friction to account for hysteresis effects.
- Validated the model through static and dynamic experiments on a physical TDCR prototype.
- Integrated a learning-based sub-model into the dynamics pipeline using PyTorch to accelerate computation and improve convergence.

Toronto, ON, Canada
May 2021 – Apr 2022

Robotics Institute, University of Toronto | Undergraduate Researcher
Supervised by Prof. Jessica Burgner-Kahrs - Continuum Robotics Laboratory

- Implemented a feedforward neural network (FNN) to learn the forward kinematics of concentric tube continuum robots (CTCRs) using PyTorch, reducing the tip position error by 81% and tip orientation error by 25% compared to state-of-the-art physics-based methods.
- Proposed and implemented a learning-based differential inverse kinematics framework for CTCRs, achieving path-following errors of 0.44 mm and 6.21 mm for simple and complex paths in simulation.

Toronto, ON, Canada
May 2019 – Apr 2020

Robotics Institute, University of Toronto | Undergraduate Researcher
Supervised by Prof. Xinyu Liu - Microfluidics and BioMEMS Laboratory

- Developed an anti-freezing, ambient-stable, and highly stretchable ionic skin with strong surface adhesion for wearable sensing and soft robotics applications.
- Characterized the ionic skin's mechanical, electrical, anti-freezing, and surface adhesion properties through extensive testing under varied environmental conditions.
- Demonstrated the piezoresistive sensing capabilities of the ionic skin by integrating it into a wearable glove to measure joint deformation and control a robotic hand via Arduino.

Industry Experience

Toronto, ON, Canada
Nov 2024 – Present

AMD | Silicon Design Engineer 2
Technology and Engineering - SerDes Technology - Applications and Validation Team

- Owned firmware development and maintenance for gigabit transceivers (GTs), delivering feature enhancements and resolving bugs in C++ in response to internal and customer feedback.
- Implemented automation and regression infrastructure across multiple GT firmware repositories, enabling automated simulation and hardware testing via GitHub Actions.
- Developed firmware unit tests for new GTs, increasing line coverage from 4.4% to 93.7% and function coverage from 13.2% to 100%, reducing regression risk and supporting safer development.
- Contributed to Vivado design tool releases and patches, delivering new features and bug fixes for the Transceivers Wizard IP to ensure software stability and readiness for customer use.

San Jose, CA, USA
Jun 2020 – May 2021

Xilinx | SerDes Application Design Intern
Wired and Wireless Group - Wired System Engineering - Applications and Validation Team

- Developed 50+ unit test cases to validate API functionality for FPGA control using C++.
- Verified tests on virtual machines and hardware prototypes across multiple communication protocols.
- Performed regression testing following hardware upgrades to ensure consistent software performance.

Publications

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| 2022 | 1. Grassmann, R. M., Chen, R. Z. , Liang, N. & Burgner-Kahrs, J. <i>A Dataset and Benchmark for Learning the Kinematics of Concentric Tube Continuum Robots</i> in <i>IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)</i> (2022), 9550–9557. |
| 2021 | 2. Ying, B., Chen, R. Z. , Zuo, R., Li, J. & Liu, X. An Anti-Freezing, Ambient-Stable and Highly Stretchable Ionic Skin with Strong Surface Adhesion for Wearable Sensing and Soft Robotics. <i>Advanced Functional Materials</i> 31 , 2104665 (2021). |

Presentations

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| 2023 | National Center for Image-Guided Therapy (NCIGT) Summer Research Retreat , Brigham and Women's Hospital
Speaker at the Continuum Robotics session, "Robotic Assisted Intracardiac Echocardiography (ICE) Catheter." |
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Honors and Awards

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| 2021 | Faculty of Applied Science and Engineering Dean's Innovation Fellowship , University of Toronto
Awarded to undergraduate engineering students conducting research in the areas of "Smart Cities, Smart Health, and The Internet of Things." |
| 2019 | Undergraduate Student Research Award , Natural Sciences and Engineering Research Council of Canada (NSERC)
Awarded to undergraduate students pursuing research in the natural sciences and engineering, based on outstanding academic record and demonstrated research aptitude. |
| 2017 | Albert and Rose Jong Entrance Scholarship , University of Toronto
Presented for demonstrating leadership in the Chinese-Canadian community as a first-year student in the Electrical Engineering or Engineering Science program. |
| 2017 | Faculty of Applied Science and Engineering Admission Scholarship , University of Toronto
Awarded for outstanding academic achievement in the prerequisite courses as a first-year engineering student. |

Technical Skills

Robotics Engineering: Robot Modeling, Machine Learning, Control Theory, Computer Vision, Human-Robot Interaction

Programming Languages: Python, C++ , C, MATLAB, HTML

Developer Tools & Workflow: Git, GitHub, Perforce, Jira, CMake, LaTeX

Libraries & Frameworks: NumPy, PyTorch, CUDA, Scikit-learn, Pandas, Matplotlib, Seaborn